

Ex. No: 10 Paging Technique

C PROGRAM

```
#include <stdio.h>

int main()
{
    int pageTable[20];
    int pageSize;
    int numPages;
    int logicalAddress;
    int pageNumber;
    int offset;
    int frameNumber;
    int physicalAddress;
    int i;

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);
    printf("Enter Number of Pages: ");
    scanf("%d", &numPages);
    printf("Enter Frame Numbers for Each Page:\n");
    for(i = 0; i < numPages; i++)
    {
        printf("Page %d -> Frame: ", i);
        scanf("%d", &pageTable[i]);
    }
    printf("Enter Logical Address: ");
```

```
scanf("%d", &logicalAddress);
pageNumber = logicalAddress / pageSize;
offset = logicalAddress % pageSize;
if(pageNumber >= numPages)
{
printf("Invalid Logical Address\n");
return 0;
}
frameNumber = pageTable[pageNumber];
physicalAddress = (frameNumber * pageSize) + offset;
printf("\nPage Number : %d", pageNumber);
printf("\nOffset : %d", offset);
printf("\nFrame Number : %d", frameNumber);
printf("\nPhysical Address : %d\n", physicalAddress);
return 0;
}
```

Output:

```
(kali㉿kali)-[~]  
$ nano paging.c  
  
(kali㉿kali)-[~]  
$ gcc paging.c -o paging  
  
(kali㉿kali)-[~]  
$ ./paging  
Enter Page Size: 100  
Enter Number of Pages: 3  
Enter Frame Numbers for Each Page:  
Page 0 → Frame: 5  
Page 1 → Frame: 2  
Page 2 → Frame: 8  
Enter Logical Address: 150  
  
Page Number : 1  
Offset : 50  
Frame Number : 2  
Physical Address : 250
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Page Size:"  
  
read pageSize  
  
echo "Enter Number of Pages:"  
  
read numPages  
  
declare -a pageTable  
  
for ((i=0;i<numPages;i++))  
do  
  
echo "Enter Frame Number for Page $i:"  
  
read pageTable[$i]  
  
done  
  
echo "Enter Logical Address:"
```

```
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
echo "Invalid Logical Address"
exit
fi
frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))
echo "Page Number : $pageNumber"
echo "Offset : $offset"
echo "Frame Number : $frameNumber"
echo "Physical Address : $physicalAddress"
```

Output:

```
(kali㉿kali)-[~]
$ bash lab10.sh
Enter Page Size:
5
Enter Number of Pages:
4
Enter Frame Number for Page 0:
5
Enter Frame Number for Page 1:
6
Enter Frame Number for Page 2:
7
Enter Frame Number for Page 3:
8
Enter Logical Address:
5
Page Number : 1
Offset : 0
Frame Number : 6
Physical Address : 30
```